



Team: *Motor Group*

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1. Progress report

Achieved since the Introduction Lecture :

- Agreed on a standard testbench design
- Designed schematics for our setup
- Collected some basic Microphone and IMU test data
- Agreed on basic and advanced data labels
- Ordered all the necessary components
- Started work on signal preprocessing
- Finished a basic Micropython script

Activities planned until next meeting:

- Record a sufficient amount of data for our models
- Design 3 AI models: IMU, Microphone, IMU + Microphone

Issues / Support needed

- IMU insensitivity
- Weird storage behaviour
- Minor Deepcraft performance issues
- Question: Training a model on two data streams from different sensors



2. Progress report

Achieved since last meeting:

- Collected a lot of data for both audio and IMU based models
- Started developing and optimizing both microphone and IMU models
- Tweaked the Micropython code to fit our current testbench setup
- Created script for improved impact data collection (IMU for now)

Activities planned until next meeting

- Optimize further for model accuracy
- Create a script for optimized impact training for the microphone
- Convert and upload the models to the board
- Combine the microphone and IMU data into a singular model

Issues / Support needed

- Why is label overlapping not allowed?
- IMU and Mic data not in sync when recording both simultaneously
- Model fails to train
- Filtering training data before model training – necessary or not?

Proposed features for Deepcraft studio:

- Add a confirmation window before recording over unsaved data
- “magnetic” snapping of labels next to each other to prevent overlap
- Feature to fill label gaps with other labels
- Add the predefined label tabs back to an already saved file
- Trimming of recorded data built in
- Continue an already existing recording
- Verbose errors saying at what time the labels are overlapping/outside data



3. Progress report

Achieved since last meeting:

- text

Activities planned until next meeting

- Text

Show what you have planned for the project presentation and your submission, you have more room on the next slide

Issues / Support needed

- text

Infineon PSOC6 Lecture presentation and final submission plans

Short project presentation:

Any special equipment need for demo at the demo Day ?